

AP Calc Lesson 2.7 Homework

Name _____

1. The table gives select values of the differentiable functions f and g and their first derivatives at $x = 2$. Let $h(x) = f(x)g(x)$. Find $h'(2)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	-4	3	2	-1

2. Let functions f and g be given by $f(x) = 3x^4$ and $g(x) = -x^2$.

a. If $h(x) = f(x) \cdot g(x)$, write an equation for $h(x)$.

b. Find $h'(x)$

c. Find $f'(x) \cdot g'(x)$

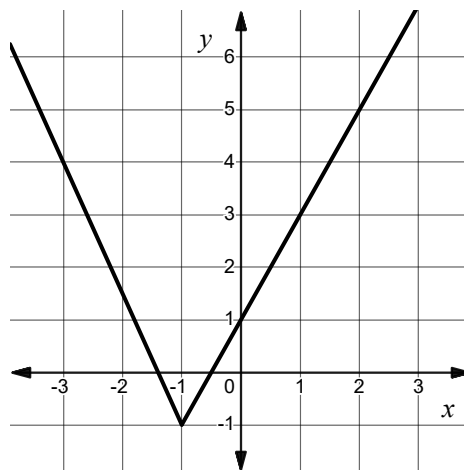
d. Explain why your answers to parts b and c are not equivalent.

3. The graph of a function $y = g(x)$ is shown in the figure. Let $j(x) = \frac{g(x)}{\sqrt{x}}$.

a. Write $j(x)$ as the product of two functions.

b. Find $j'(x)$.

c. Evaluate $j'(1)$.



4. Let $f(x) = e^x \cos x$.

a. Find the average rate of change of f on the interval $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$.

b. Find the instantaneous rate of change of f .

c. For what value of x on the interval $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$, does the instantaneous rate of change of f equal the average rate of change of f on the interval $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$?

5. The table gives selected values of a differentiable function f . Let g be the function defined by $g(x) = -f(x)(3x^2 - 5)$.

x	$f(x)$	$f'(x)$
-1	3	6
0	4	-1
1	2	7

a. Find $g'(x)$.

b. What is the value of $g'(1)$?

6. Let f be the function defined by $f(x) = (4 - 3x)e^x$.

a. Find $f'(x)$.

b. For what value of x is the line tangent to the graph of f horizontal? Show the work that leads to your solution.

7. The function f is given by $f(x) = (4x^2 - ax)g(x)$, where a is a nonzero constant and g is a differentiable function. It is known that $g(-2) = -4$ and $g'(-2) = 3$.

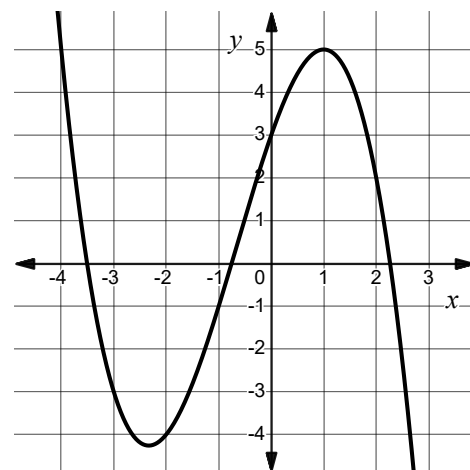
a. Find $f'(x)$ in terms of a .

b. For what value of a is $f'(-2) = 2$

8. Let f be an even, differentiable function. It is known that f is positive and increasing for all $x > 0$. The graph of a function $y = g(x)$ is given. Let $h(x) = f(x)g(x)$.

a. Which of the following statements about $h'(-1)$ is true?

- i. $h'(-1) < 0$
- ii. $h'(-1) = 0$
- iii. $h'(-1) > 0$



b. Explain your reasoning for your answer to part a.

9. Let $y = x^2 \ln x$. Find $\frac{dy}{dx}$.

A) $\frac{dy}{dx} = 2$

B) $\frac{dy}{dx} = 2x \ln x$

C) $\frac{dy}{dx} = 2(1 + x \ln x)$

D) $\frac{dy}{dx} = x(1 + 2 \ln x)$

10. What is the slope of a line tangent to the graph of $y = \sqrt{x}(3x^2 + 4)$ at the point where $x = 4$?

A) 6

B) 48

C) 61

D) 74